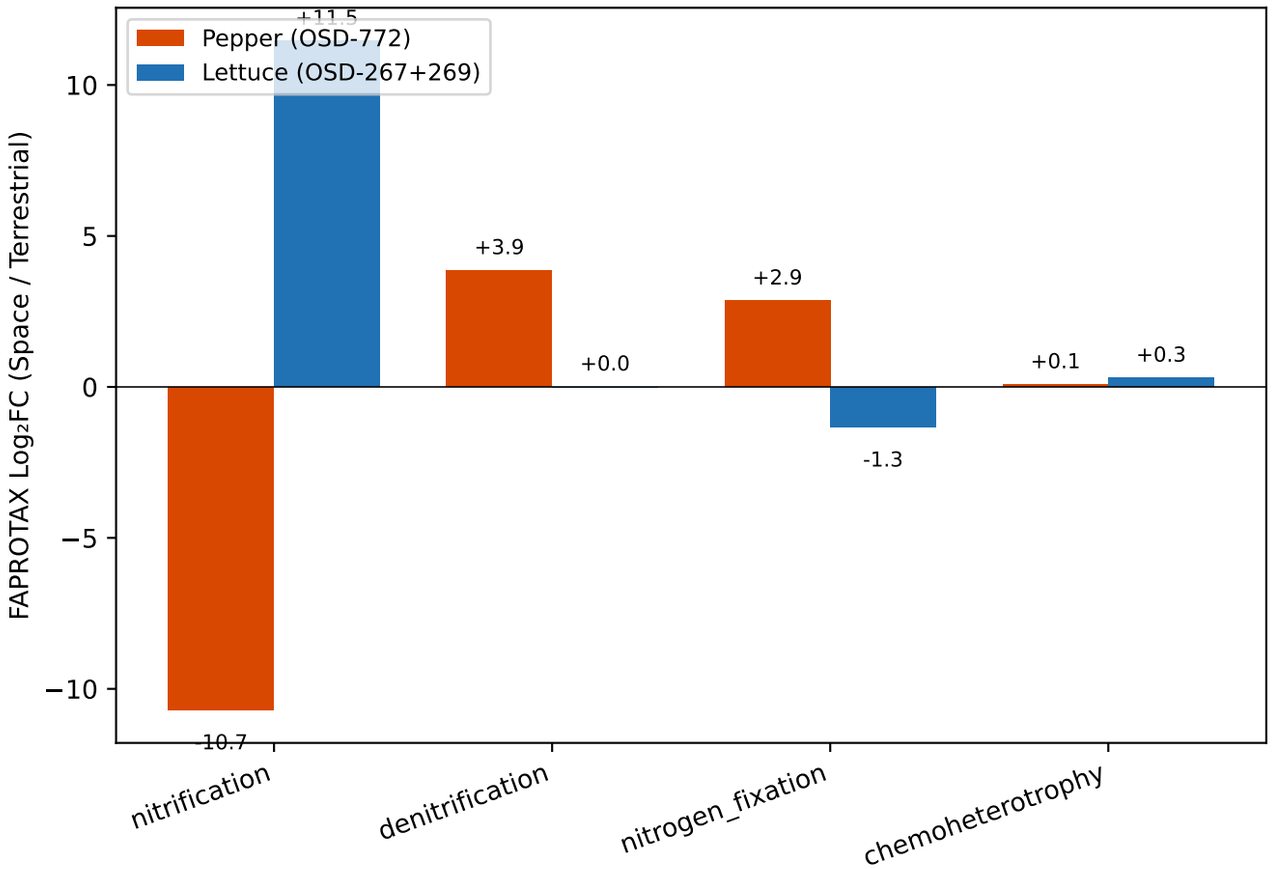
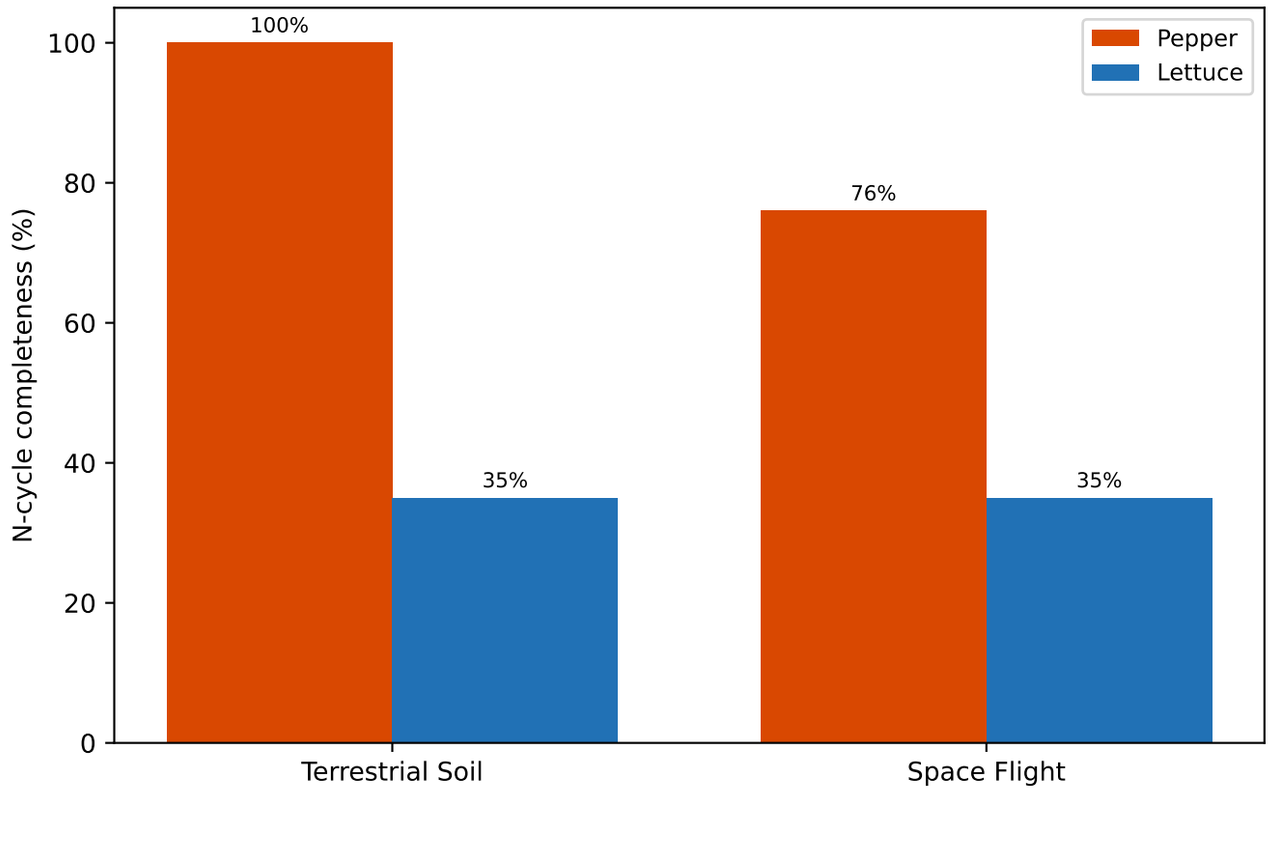
Figure 5. Crop-specific pattern — pepper and lettuce diverge

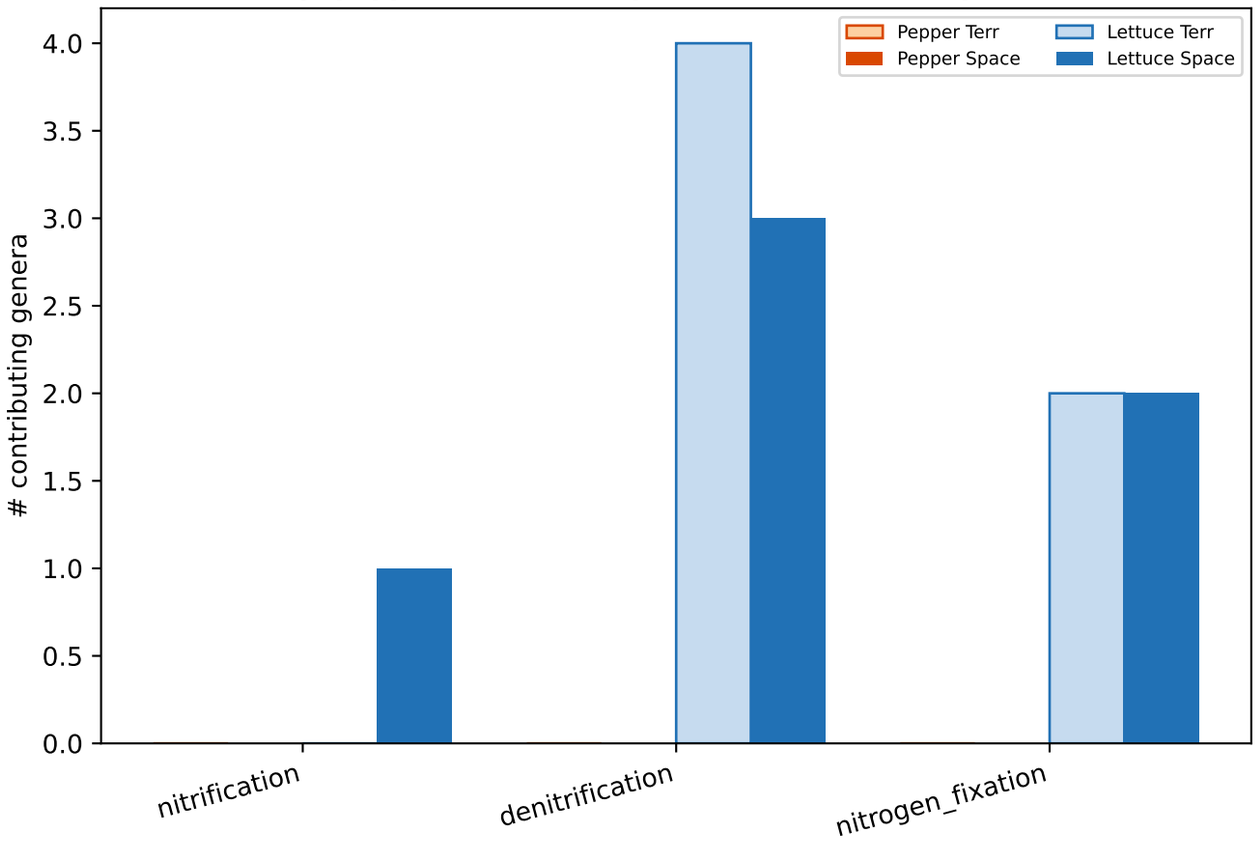
a) Divergent crop responses: pepper nitrification depletes, lettuce enriches



b) N-cycle completeness diverges between crops



c) Redundancy: pepper nitrifiers drop (4.80→0.10), lettuce pattern differs



d) Engineering implication

Crop-specific nitrification patterns:

- Pepper (OSD-772): nitrification depletion ($Log_2FC = -13.08$, 4.80 → 0.10 genera).
- Lettuce (OSD-267+269): opposite direction (nitrification $Log_2FC = +11.45$ in Space; n=4 per condition, directional only).
- Implication: BLSS biological risk models built on a single crop may not generalize. Per-crop microbiome profiling is warranted.